

Cloud and Virtualization

What the cloud really is, and why securing it is a shared job where your half is the one that breaks.

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Where we are going: seven chapters

This chapter is the foundation. The rest build on it, from securing infrastructure to real cloud attacks and the law.

1 Foundations of cloud and virtualization

2 Securing cloud infrastructure

3 Identity and access management

4 Protecting cloud data

5 Cloud network security architecture

6 Cloud threats and attacks

7 Governance, risk and compliance

Plus six lab activities and a mini project, in AWS and Azure.



True, or misconception?

Vote on each.

S1

When I move to the cloud, the provider secures my data, so I have little left to worry about.

S2

A container is really just a small, lightweight virtual machine.

S3

Most cloud breaches happen because the cloud provider was hacked.

S4

In the cloud, the network firewall at the edge is still my main line of defence.



DEFINITION

What "the cloud" actually is

The cloud is not a place, it is a model: on-demand access to a shared pool of computing resources you rent by the minute, defined by the US standard NIST SP 800-145.



On-demand self-service

Spin up a server yourself, in minutes, with no human in the loop.



Broad network access

Reach it from anywhere over the internet, from any device.



Resource pooling

Many tenants share the same physical hardware (multi-tenancy).



Rapid elasticity

Scale out under load and back down again, almost instantly.



Measured service

Everything is metered, so you pay only for what you use.

Five characteristics, one theme

You give up control of the hardware in exchange for speed and scale. Security has to move up the stack with you.

Definition: NIST Special Publication 800-145, The NIST Definition of Cloud Computing.



IaaS, PaaS, SaaS: how much you rent

The three service models differ in how much the provider runs for you. The more they run, the less you manage, and the less you can misconfigure.

SaaS

Software as a Service

You just use the app. The provider runs everything under it.

Microsoft 365, Gmail, Salesforce

PaaS

Platform as a Service

You deploy your code. The provider runs the servers and runtime.

Azure App Service, AWS Elastic Beanstalk

IaaS

Infrastructure as a Service

You rent virtual machines and networks and run the rest yourself.

*AWS EC2 (Elastic Compute Cloud),
Azure Virtual Machines*

You manage LESS



You manage MORE

Serverless (FaaS), such as AWS Lambda and Azure Functions, goes one step further: you deploy only a function, and the provider runs everything else on demand.

Models: NIST SP 800-145. Examples current as of 2025.



Public, private, hybrid, and multi-cloud



Public cloud

Shared infrastructure rented from a provider like AWS or Azure. Cheap and elastic, but multi-tenant.



Private cloud

Cloud technology run for one organisation only, on-premises or hosted. More control, more cost.



Hybrid cloud

Public and private joined together, so workloads and data can move between them.



Community / multi-cloud

Shared by several similar organisations, or spread across more than one provider to avoid lock-in.

Most real organisations are hybrid or multi-cloud, which means security has to work consistently across more than one environment at once.



PART A

How the cloud is built: virtualization

Every cloud server you rent is a virtual machine or a container sharing real hardware with strangers.



Virtualization and the hypervisor

Virtualization lets one physical machine run many isolated virtual machines. The software that creates and polices them is the hypervisor.



Type 1: bare-metal

Runs directly on the hardware. Fast and hardened. This is what powers the cloud.

Examples: VMware ESXi, Microsoft Hyper-V, KVM, Xen



Type 2: hosted

Runs as an app on top of an ordinary operating system. Great for a laptop lab.

Examples: VMware Workstation, VirtualBox

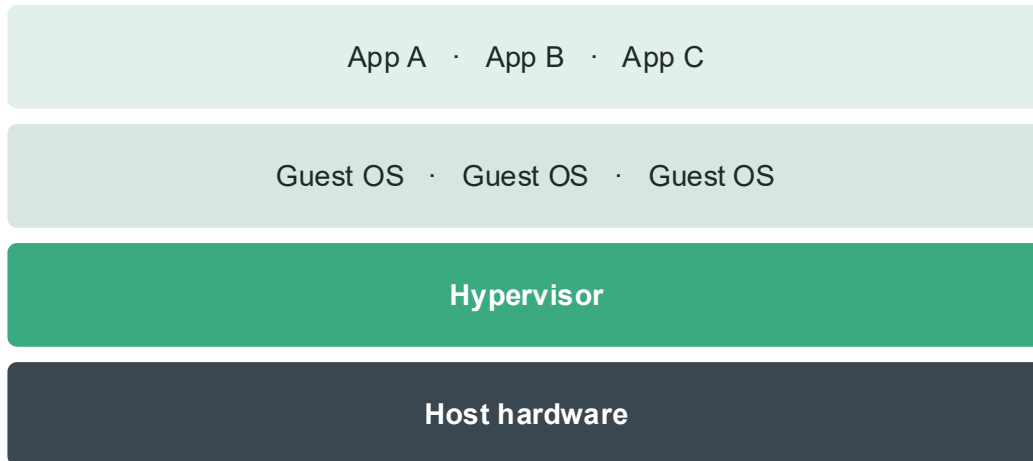


Why it matters for security: the hypervisor is the trust boundary between tenants. If an attacker breaks out of a VM into the hypervisor (a "VM escape"), every other tenant on that host is exposed. That is why the hypervisor is the most protected layer in the cloud.



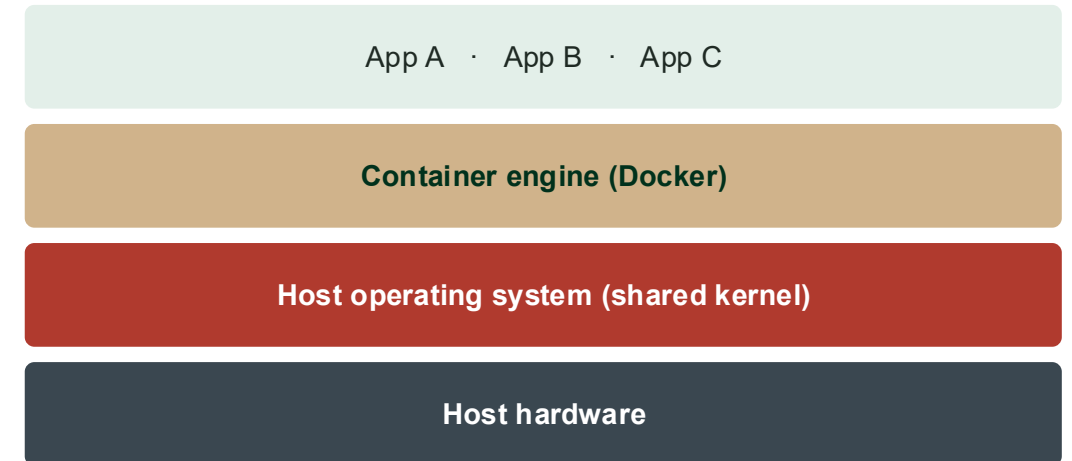
Virtual machines vs containers

Virtual machines



Each VM carries its own full OS. Strong isolation, but heavy: gigabytes and minutes to boot.

Containers



Containers share the host kernel. Light and fast: megabytes and seconds, but a thinner isolation wall.

Security consequence: a container shares the host kernel, so a kernel flaw can let an attacker "break out" of the container onto the host. This is why S2 in the warm-up is a misconception: a container is not just a small VM. Orchestrators like Kubernetes add scale, and more to secure.



Place each technology

In pairs, match each item on the left to the layer it belongs to on the right. Write the letter next to each number.

Technology

1. VMware ESXi
2. Docker
3. VirtualBox on your laptop
4. AWS EC2 virtual machine
5. Microsoft 365
6. AWS Lambda

Layer

- A. Type 1 (bare-metal) hypervisor
- B. Type 2 (hosted) hypervisor
- C. Container runtime
- D. IaaS compute
- E. SaaS application
- F. Serverless (FaaS)



PART B

Who secures what: shared responsibility

The single most important idea in cloud security. Get this wrong and you leave the door open yourself.



Security OF the cloud, and IN the cloud

Every provider draws the same line. They secure the cloud itself. You secure what you put in it.



The provider secures OF the cloud

- Physical data centres and hardware
- The hypervisor and host layer
- The global network backbone
- Managed service internals



You secure IN the cloud

- Your data, and how it is encrypted
- Identity, access and permissions
- Operating systems and patches you run
- Network and firewall configuration
- Correct configuration of every service

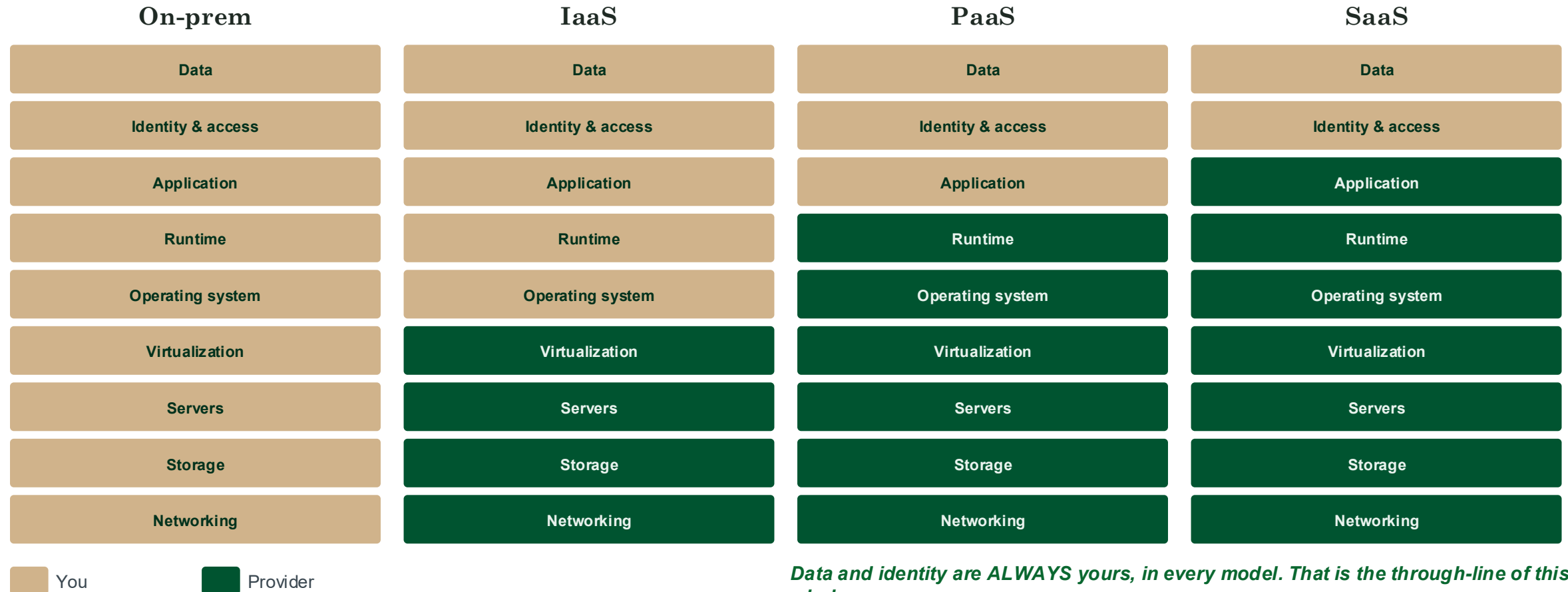
The phrase to remember: the provider gives you a secure cloud, not a secure configuration.

Framing: AWS Shared Responsibility Model; Microsoft Azure shared responsibility in the cloud.



The line moves as you move up the stack

The same nine layers, four ways to split them. Green is the provider, sand is you. Notice your share shrinks left to right, but never disappears.



Data and identity are ALWAYS yours, in every model. That is the through-line of this whole course.

Adapted from the AWS and Microsoft Azure shared responsibility matrices.



Who is responsible?

For each situation, decide who is responsible: the provider, or you. Then say which service model it assumes.

A1. Patching the physical server in the data centre.

A2. An S3 storage bucket left open to the whole internet.

A3. Patching the operating system of an EC2 virtual machine you launched.

A4. A weak password on an admin account with no MFA.

A5. A flaw inside the Microsoft 365 service code itself.

A6. Encrypting the customer records you store in the cloud.

Verdicts next. Watch for the trap: most of these are YOU.



Why cloud security is different



Multi-tenancy

You share physical hardware with strangers. Isolation is enforced by software, not walls.



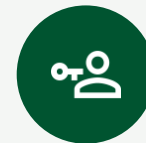
Elastic and ephemeral

Servers appear and vanish by the minute. You cannot protect what you cannot even see.



Everything is an API

The whole environment is built and changed through APIs and code, so one wrong call exposes it all.



Identity is the new perimeter

There is no network edge to hide behind. A stolen key or token is a stolen data centre.

The old perimeter is gone. In the cloud, identity and configuration are the perimeter, which answers warm-up S4.



Where cloud breaches actually come from

The Cloud Security Alliance surveyed 500+ experts for its 2024 Top Threats. The leaders are all on the customer side of the line.

1 Misconfiguration and weak change control

2 Identity and access management failures

3 Insecure interfaces and APIs

4 Weak cloud security strategy

5 Insecure third-party resources

The pattern

The provider almost never fails. The top three cloud threats are all things the customer controls: how it is configured, who can get in, and how the APIs are secured.

This answers warm-up S3.

Chapter 6 dissects a real one: the 2019 Capital One breach, where a single misconfiguration exposed 100 million records.

Source: Cloud Security Alliance, Top Threats to Cloud Computing 2024.



Your four votes, settled



Provider secures my data, little left to worry about

Misconception. You always own your data and identity. The provider secures the cloud, not your configuration.



A container is just a small virtual machine

Misconception. A container shares the host kernel, so its isolation wall is thinner. Different model, different risks.



Most breaches are the provider being hacked

Misconception. The CSA top three are misconfiguration, identity, and APIs, all on your side of the line.



The edge firewall is still my main defence

Misconception. There is no edge. Identity and configuration are the perimeter now.



Chapter 1 recap



The model Cloud is on-demand, pooled, elastic, metered computing (NIST). IaaS, PaaS and SaaS differ in how much you rent.



The engine Virtualization and the hypervisor create the cloud. VMs carry a full OS; containers share the kernel, a thinner wall.



The core idea Shared responsibility: the provider secures the cloud, you secure what is in it. Data and identity are always yours.



The reality Cloud breaches come from the customer side: misconfiguration, identity, and APIs. That is what the rest of the course fixes.



CHAPTER 1 · THE IDEA THAT RUNS THROUGH EVERYTHING

The cloud is someone else's computer. Security is shared, and your half is where it breaks.

Next: Chapter 2, securing cloud infrastructure, where we start closing your half of the door.

